EE3011.ELECTRONICS 1

Experiment (6)

Mosfet IV Characteristics

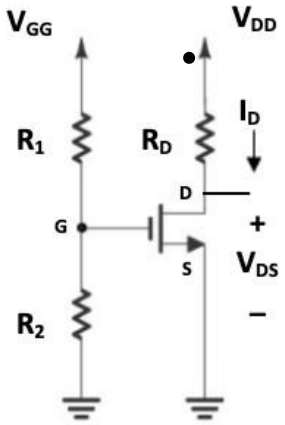
(Section A)

Mehmet Akif Türkmenoğlu

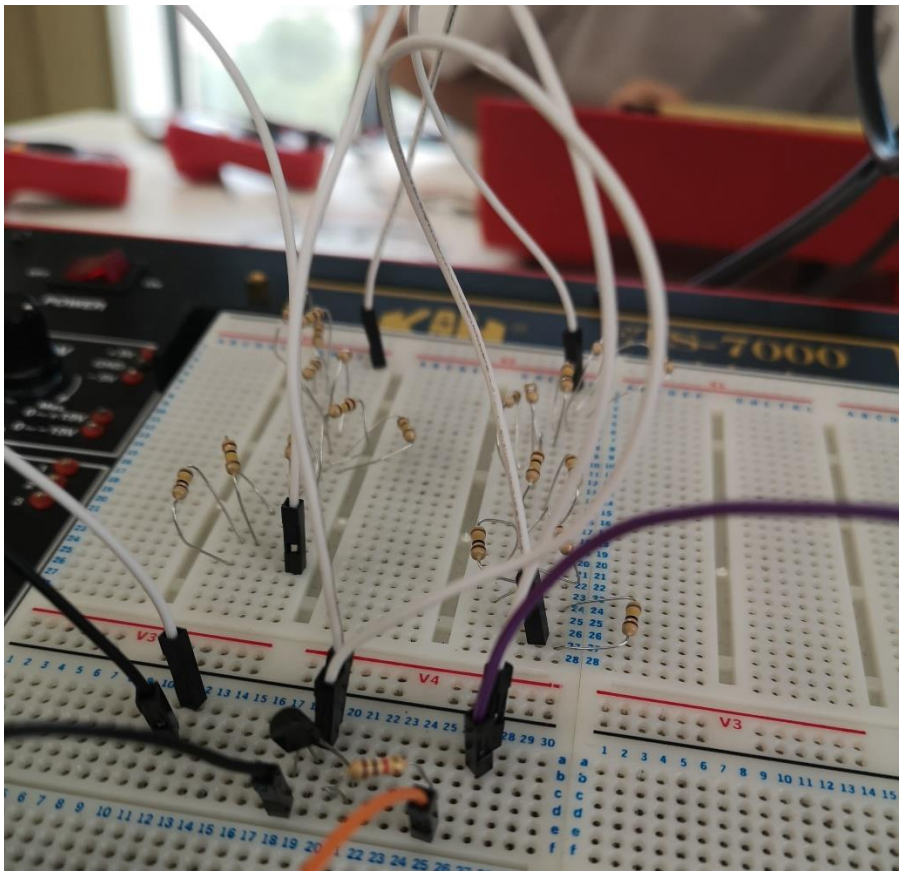
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Part (1): MOSFET IV Characteristics

For the first part, I connected to circuit as it is and give the values of 1Mega Ohm to R1 and R2, 1k to RD and +8V to VGG. Then I change the voltage of VDD as starting from 0 to 10V.



This is the circuit that I have.

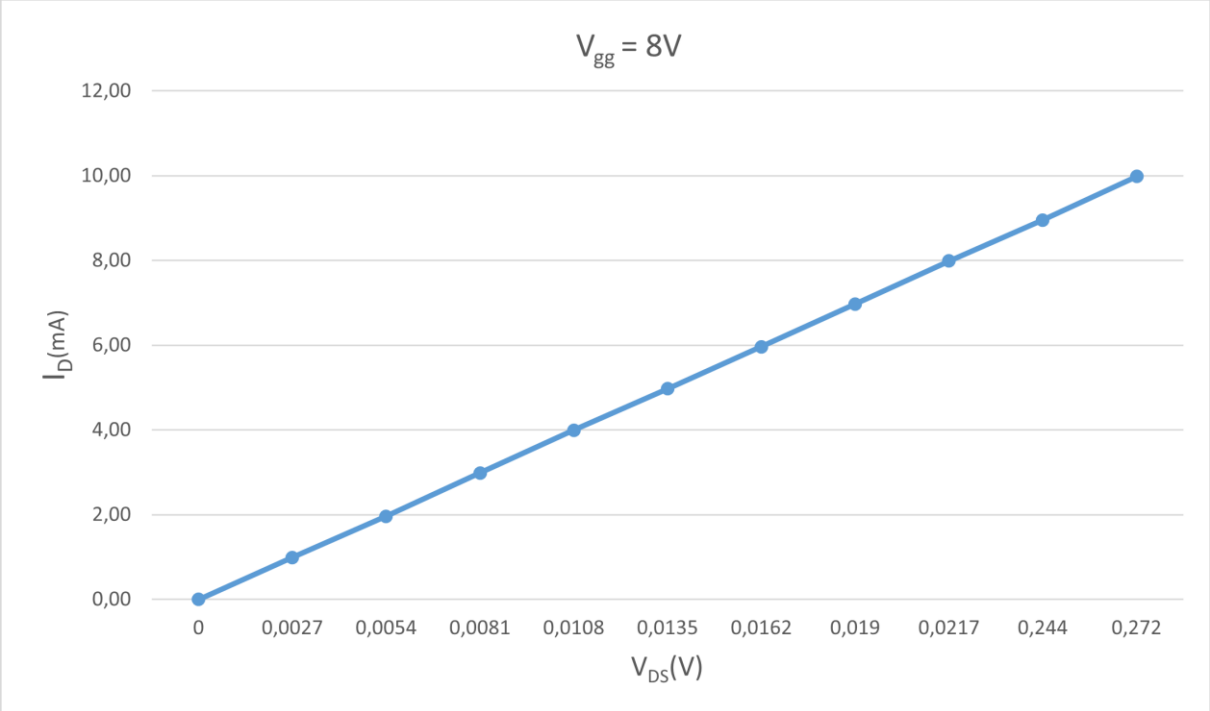


Because there is no 1 MegaΩ resistor I connected 10 10kΩ resistors to get 1 MegaΩ. We use this much resistor to ignore the value. Then I record the voltage and current values and merge them in table 2 and graph 1.

Table 2. IV Characteristics Measurements

	$V_{GG} = 8V, V_{GS} = 4V$			$V_{GG} = 10V, V_{GS} = 5V$		
V_{DD} (V)	V_{DS} (V)	V_{R_D} (V)	I_D (mA)	V_{DS} (V)	V_{R_D} (V)	I_D (mA)
0.00	0	0.001	0.001	0	0.001	0.001
1.00	0.0027	0.99	0.99	0.0021	0.897	0.897
2.00	0.0054	1.99	1.99	0.0043	1.93	1.93
3.00	0.0081	2.99	2.99	0.0064	2.91	2.91
4.00	0.0108	3.98	3.98	0.0085	3.90	3.90
5.00	0.0135	4.99	4.99	0.0106	4.91	4.91
6.00	0.0162	5.98	5.98	0.0128	5.90	5.90
7.00	0.0190	6.98	6.98	0.0149	6.93	6.93
8.00	0.0217	7.97	7.97	0.0170	7.92	7.92
9.00	0.244	8.94	8.94	0.0192	8.86	8.86
10.00	0.272	9.97	9.97	0.0214	9.91	9.91

Graph 1. Mosfet IV Curves



1. Comment on the IV curves that you obtained. Can you identify the transistor modes ofoperation?

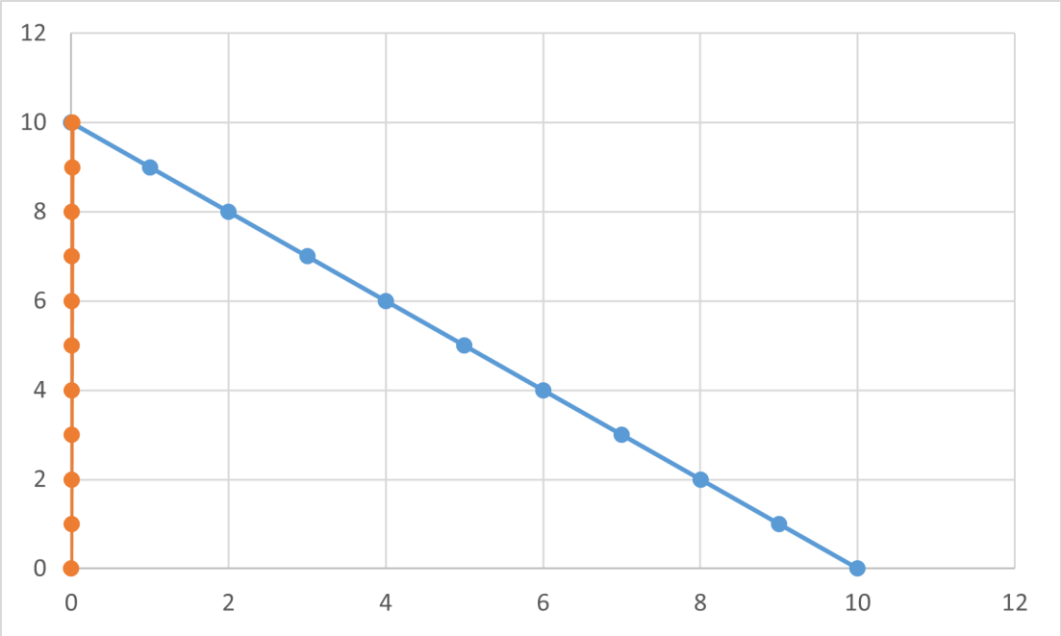
Answer: The mosfet is in saturation mode and the IV curve is linearly increasing.

2. Is the drain current constant in the saturation region of the IV curve? Explain why or whynot.

Answer: Drain current is not constant it changes according to r0.

Part (2): Graphical Analysis of a DC MOSFET Circuit.

We asked to draw the load line on top of the IV curves in graph 1. I marked the intersection point of the curves.



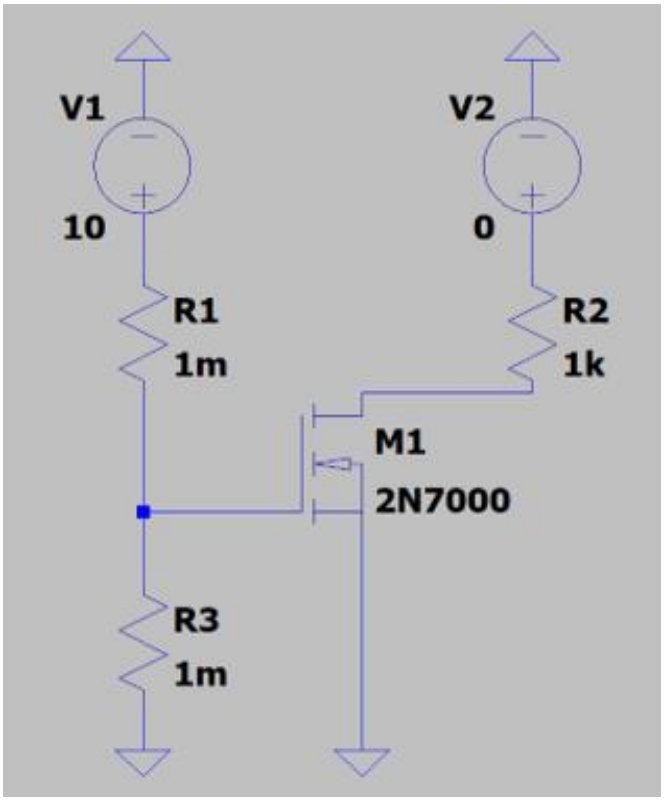
The experimental value of $I_D = (V_{DD} - V_{DS})/R_D$.

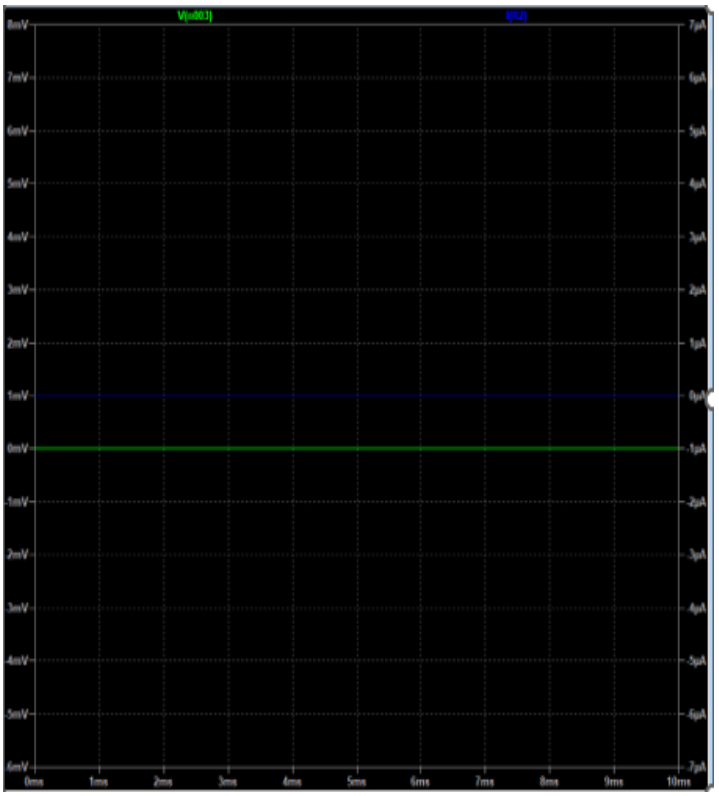
Then I record the results in table 3.

Table 3. Graphical Solution Results.

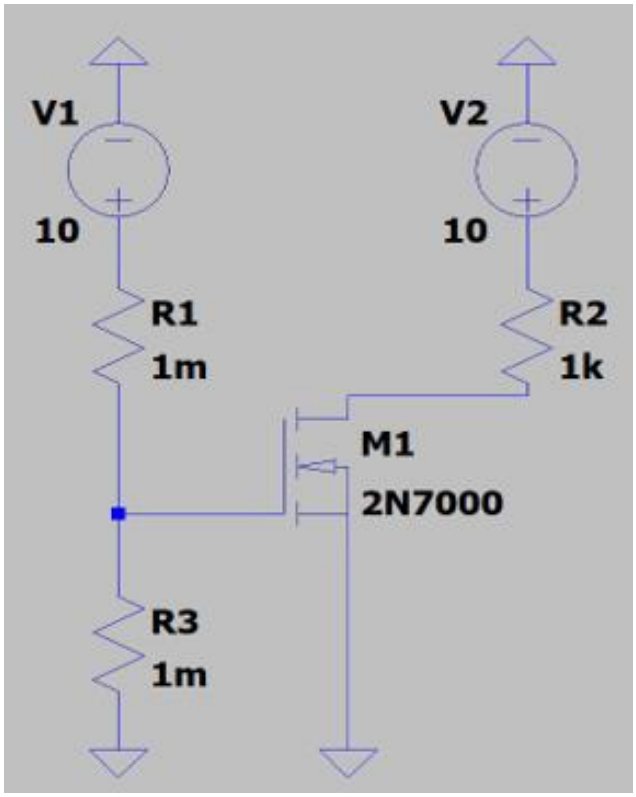
V_{DD}= 10 V and V_{GG} = 10 V					
V_{DS} (V) Experimental value	V_{DS} (V) Graphical value	% Difference	I_D (mA) Experimental value	I_D (mA) Graphical value	% Difference
0,024	0,026	7,6923	10,01	9,97	0,4011

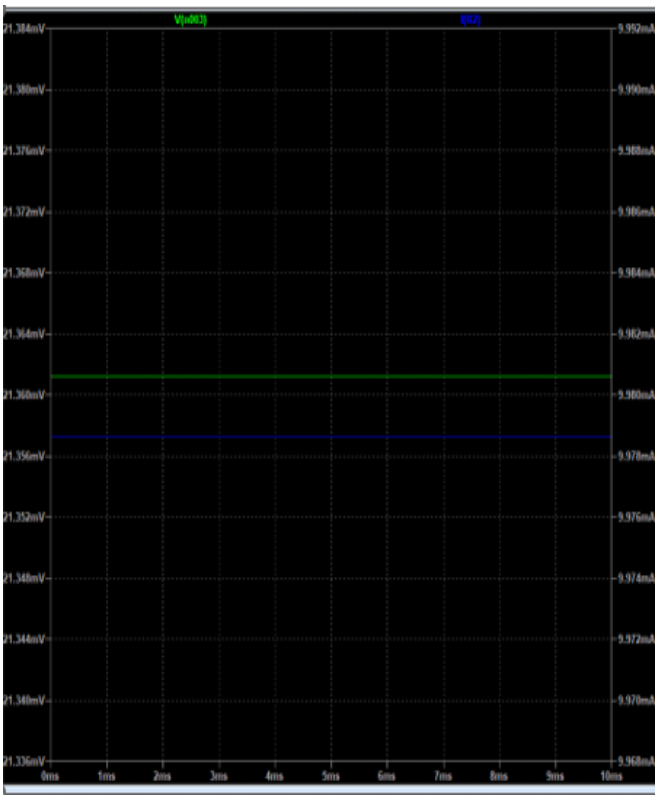
If we increase the voltage value more slowly, the error decreases.



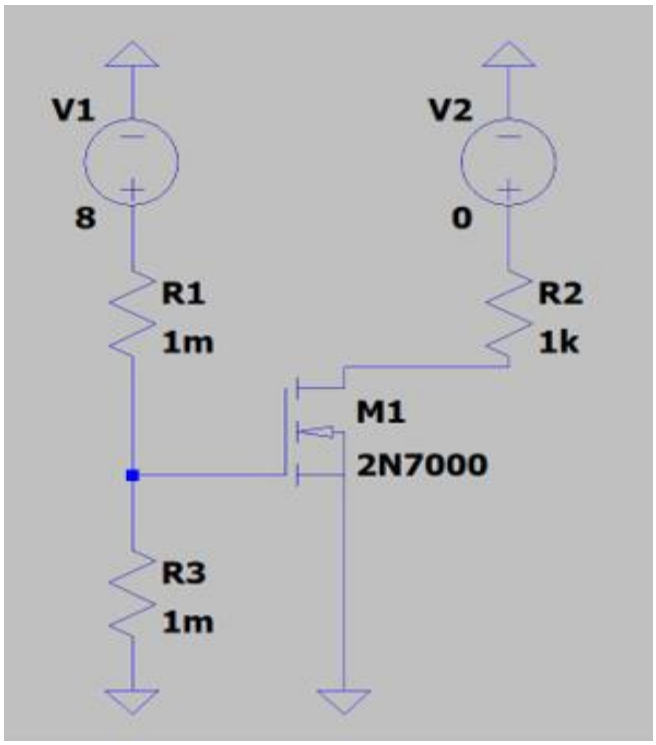


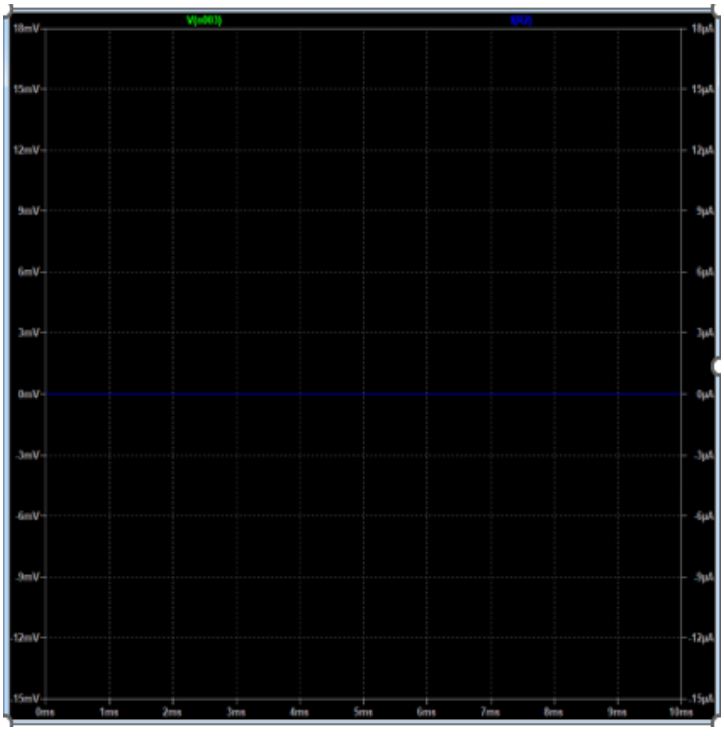
V_{DD}= 0 V and V_{GG} = 10 V



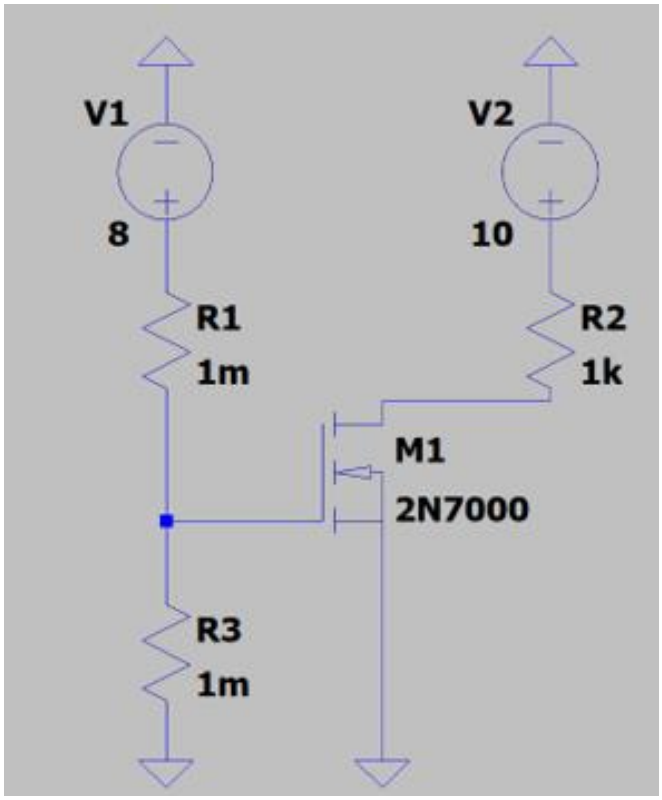


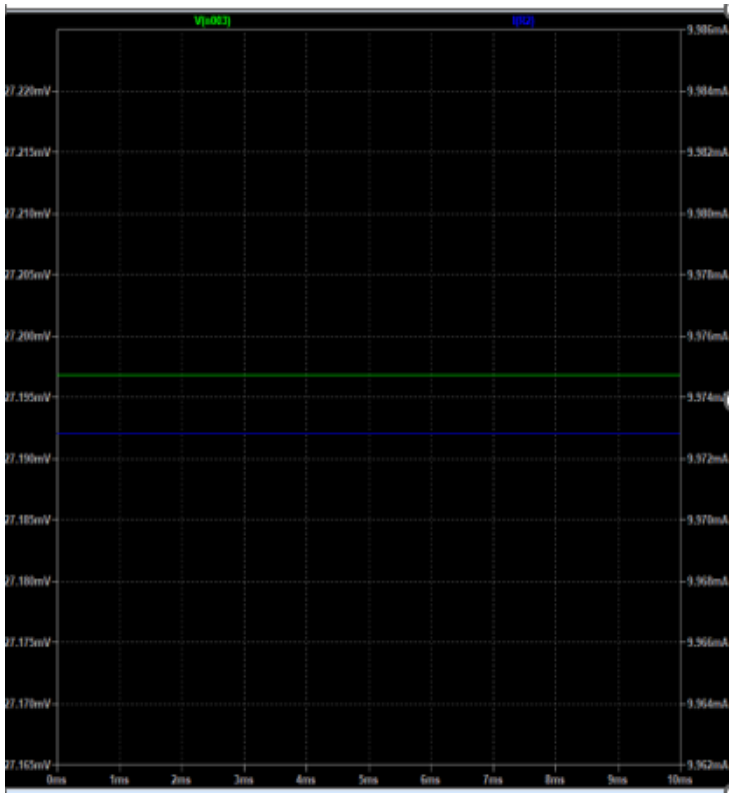
V_{DD}= 10 V and V_{GG} = 10 V





V_{DD}= 0 V and V_{GG} = 8 V





V_{DD}= 10 V and V_{GG} = 8 V

These are the Ltspice simulations. I don't need to put all the experimental values. I am sure of my measurements and calculations. I hope it is correct.